Wildfire Segmentation using Deep-RegSeg Semantic Segmentation Architecture

Rafik Ghali
Moulay A. Akhloufi
{rafik.ghali,moulay.akhloufi}@umoncton.ca
Perception, Robotics, and Intelligent Machines Research Group (PRIME), Department of Computer Science, Université de Moncton, Moncton, NB
Canada

Wided Souidene Mseddi
Marwa Jmal
wided.souidene@ept.rnu.tn;jmal.marwa@gmail.com
SERCOM Laboratory, Ecole Polytechnique de Tunisie, Université de Carthage, La Marsa
Tunisie

Abstract

Wildfires are a worldwide natural risk, which causes harmful effects to human safety and leads to ecological and economical damage. Various fire detection systems have been proposed in order to detect fire and reduce its effects. However, they are still limited in detecting small fire areas and determining the precise fire's shape. In order to overcome these limitations, we present, in this paper, a novel method based on deep learning, called 'Deep-RegSeg', to segment fire pixels and detect fire areas in complex non-structured environments. Deep-RegSeg is evaluated with varying backbone and loss function. The obtained results showed a high performance and outperformed some recent state-of-the-art techniques. The results also proved that Deep-RegSeg is efficient in segmenting wildfire pixels and detecting the precise fire's shape, especially small fire areas under various conditions of weather, presence of smoke, and environment brightness.

Keywords

Fire segmentation, Deep-RegSeg, Forest fire, Deep Learning

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1 Introduction

Wildfires are natural disasters, which are caused by natural phenomena and/or human activities. They have immediate and long term harmful effects on human safety, ecological and economical balance, air pollution, etc. Early fire detection systems are developed to improve fire fighting and reduce their disastrous effects, their intensity, and size. The earliest detection systems used fire sensing technologies such as gas, heat, and flame sensors. However, these systems are still limited by coverage areas, slow detection time, and false alarms [13]. To avoid these limitations, researchers adopted vision sensors for early wildfire detection by recognizing color, shape, texture, and motion features of fire. These sensors have a great success thanks to their low cost and their large coverage areas [13].

Recently, Deep Learning (DL) methods improved computer vision tasks to solve challenging problems such as medical image analysis, satellite image analysis, video recognition, person tracking, motion estimation, and object detection and segmentation [24]. DL methods also achieved an excellent performance compared to hand-craft learning methods in various tasks [24]. Due to their great performance, DL approaches are used in fire detection task to detect flame and draw the bounding box of fire area [18, 25]. However, they are still limited by the wide variety of the fire's shape. Moreover, fire segmentation based on deep learning is more suitable to segment fire pixels and detect fire areas present in the images/videos [17]. Nonetheless, they are still limited in identifying the precise fire's shape as well as small fire areas due to the presence of smoke and the complexity of fire characteristics' and its changing shape.

In this paper, we use a newly introduced segmentation method based on deep learning, called 'Deep-RegSeg', to segment fire pixels and detect the precise fire's shape. Firstly, we evaluate our method under different settings and varying backbones, two loss functions that are Dice loss and Binary Cross Entropy Dice loss. Then, we compare its performance in wildfire segmentation task with U-Net, color space fusion method [10], and U^2-Net, which showed excellent results in object segmentation [15, 16].

More specifically, we present in this paper an architecture able to segment fire pixels, detect the precise shape of fire areas, and avoid the limitations of some recent state-of-the-art methods. Then, we prove the robustness of our method to segment fire pixels under various conditions of weather, environment brightness, and presence of smoke.

2 Related work

Numerous fire segmentation methods have been proposed in the literature. Feature-based fire segmentation methods are the first methods used to segment fire pixels and identify the visual features such as color, texture, and motion. At first, Color space methods such as RGB (Red, Green, Blue) [9], HSI (Hue, Saturation, Intensity) [20], and YCbCr [36] have been employed to segment fire pixels. However, they generate high false alarm rates due to their

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sensitivity to illumination changes and the difficulty of determining the exact range of the color of fire pixels. Next, color space methods have been combined to extract the characteristics of fire [10]. Then, various features, which are spatial, static and dynamic texture, motion, temporal, and color information have been exploited to segment fire pixels and detect fire areas from images and/or videos [5, 8, 11, 12, 22, 33]. These methods showed a great performance and proved their ability to reduce the false alarm rate. However, they are still limited by the detection of fire-like objects, specially in the case of real fires as well as small fire areas [14]. Recently, Deep Learning has been used for fire segmentation thanks to its great success in various challenging problems. DL proved its ability in segmenting fire pixels and detecting fire areas. For instance, Gonzalez *et al.*[1] proposed a novel approach based on CNN (Convolutional Neural Networks), SFEwAN-SD (Simple Feature Extraction with FCN AlexNet, Single Deconvolution), to monitor wildfire and estimate its spread and its location using an UAV (Unmanned Aerial Vehicle) monitoring system [2]. At first, an adaptation of AlexNet [23] is employed to determine the texture and the shape of fire. Then, a simple CNN, which contains numerous 3*3 convolution blocks followed by ReLU (Rectified-Linear Unit) activation function extracts color and texture of fire and corrects the false positive rate of AlexNet. Experimental results proved the performance of the SFEwAN-SD model to detect the fire region without generating a false alarm of fire-like objects such as sunset. Muhammad *et al.* [26] proposed a deep learning method based on the SqueezeNet [21] to detect and monitor fire using surveillance systems. This method showed an efficient performance to identify flame and detect its location. It also reduced the false alarm and minimized fire damage. It proved its ability to localize and identify fire in closed-circuit television surveillance systems for both outdoor and indoor environments. Akhloufi *et al.* [3] proposed a Deep-Fire model based on the Encoder-Decoder U-Net [29] to segment fire pixels in a non structured environment. This model achieved an accuracy of 97.09 % and an F1-score of 91 % using the CorsicanFire dataset [31] and Dice loss as loss function. It showed its ability to segment wildfire pixels and detect fire areas. Bochkov *et al.* [4] proposed a novel AI solution based on U-Net for multiclass fire segmentation. The authors illustrated a novel deep learning model, wUUNet (wide-UUNet concatenative), which consists of the refinement of U-Net with maximum skip connection's number. The UUNET concatenative architecture contains two U-Net architectures. The first architecture provides binary fire segmentation. It determines the fire area. The second U-Net provides a multi class fire segmentation. It extracts the fire's colors, which are red, orange, and yellow. Using a custom dataset of 36 videos and 6250 images and the backbone VGG16 [30]. This solution showed an excellent performance, outperforming the state-of-the-art U-Net and proved its ability to segment fire in robotic firefighting systems [4]. Mlích *et al.* exploited the winner in ILSVRC Scene Parsing Challenge 2016, DeepLabV3 [7] to localize and segment fire. Using a dataset of non-fire images from the SUN397 dataset [34] and 1775 fire images, DeepLabV3 reached a great performance outperforming the state-of-the-art models due to its multi-scale extracted feature map. It proved its high performance in segmenting fire pixels in challenging scenes for example sunset, night street lighting, and sunrise. Harkat *et al.* [19] presented a semantic segmentation method based on DeeplabV3+ for forest fire segmentation. DeeplabV3+ is trained under different settings and backbones, which are ResNet-18 and ResNet-50, loss functions that are Tversky loss, Cross Entropy loss, and Dice loss, and learning rate values of 10^{-3}, 10^{-2}, and 10^{-1}. This model reached an excellent performance on the CorsicanFire dataset [31], which consists of 1135 RGB images and 640 infrared images. It outperformed a CNN, Sn-dcnet [19], which is a simple CNN architecture with dilated convolution and proved its ability in segmenting fire pixels and detecting fire areas. Ghali *et al.* [16] adopted vision transformers (TransUNet [6] and MedT [32]) in the context of wildfire segmentation using RGB images. They demonstrated the potential of vision transformers to outperform deep CNN models. Ghali *et al.* [17] also employed vision transformers (TransUNet and TransFire) for UAV Forest fire image segmentation task. Experiment results showed high performances outperforming recent published work.

3 Proposed method

In this paper, we present a new method, called Deep-RegSeg, which adopts RegNet [28] model as a backbone to encode the features of fire. The RegNet model shows an excellent ability to extract features from images. It reaches a great performance and speed with up to 5x faster than popular EfficientNet models [28]. It also outperforms the state-of-the-art models such as MobileNetv2, ShuffleNetv2, NasNet-A, and EfficientNet models [28]. As presented in Figure 1, the proposed Deep-RegSeg architecture, is an Encoder-Decoder structure with four concatenation shortcuts. The Encoder employs RegNet model as a backbone to extract fire's features. Then, the Decoder up samples the extracted feature map to input resolution and determines the binary mask as output. The encoder consists of five blocks, Stem, stage1, stage2, stage3, and stage4. The input image is fed to the Stem block, which comprises a 3*3 convolution layer with a stride-two and output channels value of 32, followed by ReLU activation function and batch normalization layer. Then, the extracted features are generated by four stages, which reduce progressively the feature map's resolution. Each stage includes convolutional block, which uses a convolutional layer with stride two and a sequence of identical blocks (ID). We employ RegNetX as a backbone. The ID are based on the standard residual bottleneck block with group convolution [35]. Each block consists of a 1*1 convolutional layer followed by a group of 3*3 convolutional layers, and a final 1*1 convolutional layer. Each convolutional layer follows by batch normalization layer and ReLU activation function. Six parameters that are depth d, initial width w_0, slope w_a, w_m, which controls quantization, group width g, and bottleneck ratio b are employed to generate width and depth of each stage and define the RegNetX models. The Decoder consists of three Deconv blocks followed by Conv block and 1*1 convolutional layer. Each Deconv Block comprises a transposed convolution layer, followed by ReLU activation and batch normalization layer. Conv block contains a 3*3 convolution layer with stride-one and output channel value of 32, followed by ReLU activation and batch normalization layer. Finally, a 1*1 convolutional layer generates a binary mask, result of the fire segmentation process.

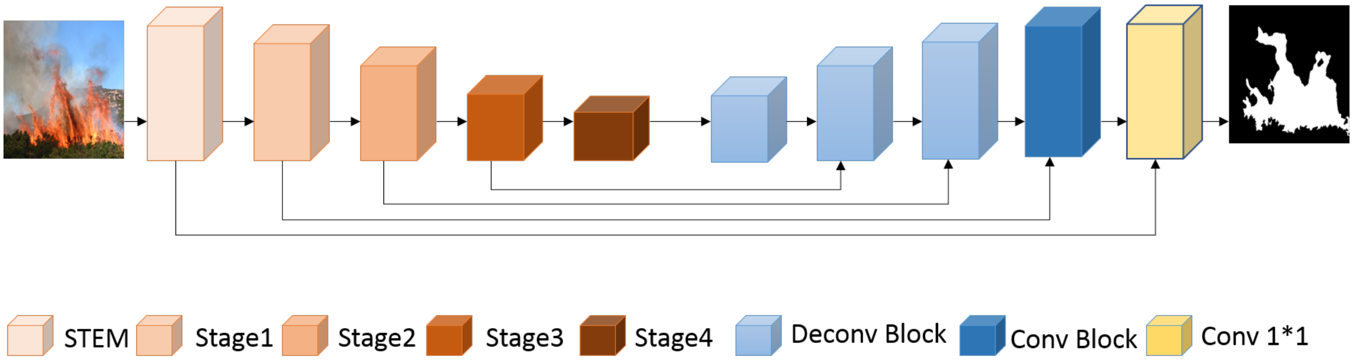

Figure 1: The Deep-RegSeg architecture.

4 Implementation and experimental results

In this section, we present the dataset used for learning and the evaluation metrics. Then, we illustrate the implementation details. Finally, we discuss the experimental results.

4.1 Dataset

To learn and test our method, we use the CorsicanFire dataset [31], which is composed of 1135 RGB wildfire images with their corresponding binary mask. The RGB images describe the fire's features such as color (Red, Orange, and White Yellow) under different conditions: varying luminosity, the environment brightness, and the presence of smoke.

4.2 Evaluation metrics

We evaluate our proposed method using the three following metrics: (1) *Accuracy* is the proportion of the correct predictions over the total predictions' number, (2) *F1-score* combines of precision and recall metrics to determine the model's performance, and (3) *Mean Test Time* is the average time of segmentation using our test data.

4.3 Experimental results and discussion

We develop our method using Pytorch [27] on a machine with NVIDIA Tesla T4 GPU. Our dataset was split randomly into three sets: 815 images for training, 111 images for validation, and 209 images for testing. Deep-RegSeg is trained from scratch (no pre-training) using horizontal flip and rotation of 15 degrees as data augmentation techniques. In addition, we used Dice loss (DC) and Binary Cross Entropy Dice loss (DBCE) as loss function to determine the similarity between ground truth and predicted images.

To evaluate the performance of our method, we first show the ability of Deep-RegSeg under different settings: varying backbones, 256*256 input resolution, and two loss functions, DC and DBCE. Then, we compare the Deep-RegSeg performance with some recent models for fire segmentation, color space fusion method [10], U-Net, and U^2-Net [15, 16]. Table 1 illustrates the experimental results of Deep-RegSeg with RegNetX models as backbone and two loss functions, Dice loss and DBCE loss. We can see that the results are very interesting and proved that the proposed deep learning model, which adopt the Encoder-Decoder structure is suitable in wildfire segmentation. Deep-RegSeg with RegNetX reaches an excellent performance thanks to its very rich extracted feature maps. Deep-RegSeg proves its ability in segmenting fire pixels and detecting exact fire areas at an early stage. Using DBCE loss, Deep-RegSeg shows better performance than Deep-RegSeg with Dice loss. Also, Deep-RegSeg with RegNetX_800MF reveals better performance with an accuracy of 94.82 % and an F1-score of 94.46 %. Deep-RegSeg with RegNetX_200MF, RegNetX_400MF, and RegNetX_800MF also obtain high results and a test time at least 8x faster than Deep-RegSeg with RegNetX_16GF and RegNetX_32GF. Based on the trade-off between performance (accuracy and F1-score) and speed (mean test time), Deep-RegSeg with RegNetX_800MF and DBCE loss obtains better results with an accuracy of 94.82 %, an F1-score of 94.46 %, and a mean test time of 0.16 second. This model also proves its ability in segmenting fire pixels and detecting exact fire areas under different conditions such as the presence of smoke and changing luminosity. As an example, we can see in Figure 2 that our proposed methods, Deep-RegSeg with RegNetX_800MF correctly distinguish between fire and non-fire pixels. Deep-RegSeg segments very well fire pixels closely to ground truth and detects exact fire areas, especially small fire areas under different conditions such as the presence of smoke and changing luminosity. Table 2 presents a comparative analysis of our method, Deep-RegSeg with recent state-of-the-art models, U-Net, color space fusion method, and U^2-Net using CorsicanFire dataset. We can see that Deep-RegSeg with RegNetX_800MF shows a better performance with an F1-score of 94.46 %. It proves its ability in segmenting fire pixels and detecting the precise shape of flame under various conditions of weather, environment brightness, and presence of smoke. In addition, we evaluate Deep-RegSeg using images downloaded from web. In Figure 3, we can see that the result of Deep-RegSeg with RegNetX_800MF is very close to the shape of the fire in the RGB input images. In addition, our proposed model is robust to changing luminosity and the presence of smoke. This model shows its ability in segmenting wildfire pixels and detecting the precise fire's shape, especially small fire areas.

To summarize, our proposed method, Deep-RegSeg, correctly distinguishes between fire and non-fire pixels. Deep-RegSeg also outperforms recent published models, U-Net, fusion color space methods, and U^2-Net.

Table 1: Comparative analysis of Deep-RegSeg using DC and DBCE loss and RegNetX models as backbone.

Backbone	Accuracy (%)	F1-score (%)	Mean Test Time (s)	Loss function
RegnetX_200MF	94.77	94.32	0.14	Dice Loss
RegNetX_200MF	94.65	94.09	**0.13**	DBCE Loss
RegNetX_400MF	91.43	89.20	0.15	Dice Loss
RegNetX_400MF	94.41	94.00	0.14	DBCE Loss
RegNetX_800MF	92.25	90.47	0.16	Dice Loss
RegNetX_800MF	**94.82**	**94.46**	0.16	DBCE Loss
RegNetX_16GF	94.66	94.29	1.22	Dice Loss
RegNetX_16GF	94.72	94.38	1.22	DBCE Loss
RegNetX_32GF	94.62	94.30	1.26	Dice Loss
RegNetX_32GF	94.67	94.06	1.43	DBCE Loss

Table 2: Comparative analysis of Deep-RegSeg, U^2-Net, U-Net, and a color space fusion methods using CorsicanFire dataset.

Models	F1-score (%)	Learning Data
Deep-RegSeg-RegNetX_800MF (DBCE loss)	**94.46**	1135 images
U-Net (DC loss) [3]	91.00	419 images
Color space fusion method [10]	79.00	500 images
U^2-Net (DBCE loss)	92.00	1135 images
U-Net (DBCE loss)	94.00	1135 images

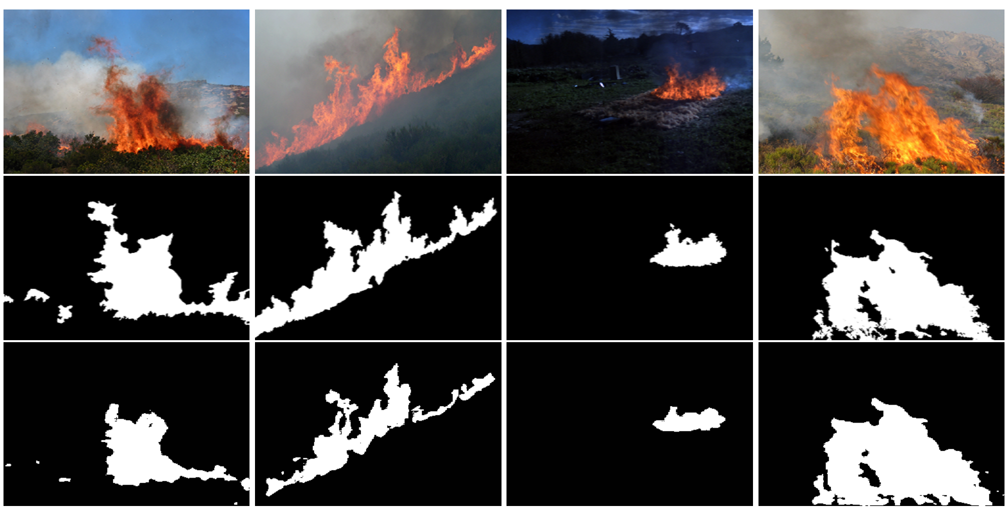

Figure 2: Deep-RegSeg results, from top to down: original images, ground truth images, and Deep-RegSeg-RegNetX_800MF results.

5 Conclusion

In this paper, we present a novel Encoder-Decoder Semantic Segmentation method, Deep-RegSeg, for wildfire segmentation. We evaluate the performance of our proposed model by varying backbone and two loss functions, DC and DBCE loss. Using CorsicanFire dataset and three evaluations metrics that are F1-score, accuracy, and mean test time, Deep-RegSeg shows an excellent performance and outperforms some recent state-of-the-art models, U-Net, fusion color space methods, and U^2-Net. It proves its ability in segmenting fire pixels and detecting fire areas, especially small fire areas under various conditions of weather, environment brightness, and presence of smoke. For future work, we aim to present a smoke/fire segmentation method based on Deep-RegSeg in order to segment both fire and smoke pixels and detect smoke/fire areas. We will also test attention mechanisms in our proposed model.

Figure 3: Deep-RegSeg results using internet collected image, from top to down: original images and Deep-RegSeg-RegNetX_800MF results.

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